

ActInfLab 2022 Quarterly Roundtable #1

Roundtable | 1:14:05

Thank you.

Hey everyone. Thanks for tuning in and today it is March 30th, 2022. We're here in the ACTIMF Lab quarterly roundtable number one for 2022. Welcome to the ACTIMF Lab. We're a participatory online lab that is communicating, learning, and practicing applied active inference. You can find us at some of the links here on this slide. This is a recorded and archived live stream, so please provide us with feedback so that we can improve our work.

All backgrounds and perspectives are welcome here and we'll be following video etiquette for live streams.

Today in the ACTIMF Lab quarterly roundtable number one for 2022, we're going to have a first section after our intros on lab scale updates.

And then we'll talk about project scale updates, which are nested within the lab scale. And just close with some preferences and expectations. And through and through, we hope that you'll see many affordances for participation.

Many ostensive cues, signals in your epistemic niche that elicit action, that make you curious about something, that connect with your pre-existing preferences or update them so that you get involved with active inference or possibly even the lab.

So whenever you do see this or listen to it or some other combination, feel free to go to activeinference.org or email us at activeinference@gmail.com and we'll be in touch from there.

So we can begin with just some introductions and warmups and it feels so different to do this without a paper that we've read before.

So for these intros, we can just say hi and anyone can reflect on just like, how is your generative model updated this quarter?

What did you perceive or learn? How did you change? And what might be something that you're looking forward to as we carry on the latter 75% of 2022?

So I'm Daniel and a researcher in California. I think my generative model updated and fragmented into a thousand little pieces with Coda and the way that we used formal documents this quarter, as opposed to previous quarters, but it's cohering in a way that is actually helpful.

I hope so. I'll pass to Alex.

Alex.

Yes. Hi, everyone. I'm Alex Svetkin. I'm one of the founders of the lab and also researcher at Systems Management School.

And from latest time, I feel like we're sharing our generative model more and more and hopefully it will bring us to new developments and they pass it to blue.

Hi, everyone. I'm blue. I am an independent research consultant in New Mexico.

I think this quarter was pretty special. Like the Coda restructuring was pretty awesome and just contributing to building like lab infrastructure in Coda and in like YouTube with like videos and timestamps and podcasts.

And I don't know, it was, it was fun. I think probably the biggest update was, was the quantum paper because that was pretty, pretty epic.

And I think the one that we just have discussed over the last couple of weeks.

And in the next quarter, I'm looking forward to having some more quiet time, less talking, more doing.

That's what I'm looking forward to in Q2 of 2022. And I will pass it to Ivan.

I'm going to pass it to Ivan.

I'm going to pass it to Ivan.

Thanks, Blue.

Yeah, it's very nice. I'm, I'm, I'm, I'm based in, uh, uh, and, uh, and, uh, and we finally run system thinking study group this, uh, this quarter. And I hope, uh, lab participants enjoy it and I see it that they do.

And I hope we run next, uh, another study group and it will be on inference. So that it, uh, I asked it to Dave.

Yeah, a lot going on. Um, I'm becoming a little less guilty about skipping at least half of the activities, but I get to hear, uh,

at least, at least to read the papers and listen to the live streams. And there's lots of work going on with the transcript, which is very gratifying.

Even if Daniel no longer believes himself about how valuable it is to do the print transcripts and put them in, in service.

I still am sticking with the 10,000 times more value from a transcribed talk than an, just a YouTube audio with pictures.

So who haven't we heard from Steven?

Hello. Yeah, we're still here. We're still standing sort of, so that's great. I'm here in Toronto. Um, for me, I'm, I'm appreciative for certain of the capacity, the labs offered, um, the connections, the conversations, the ability to bounce the ideas.

And that's been really, really helpful. And, um, particularly with like Mark's, uh, soames and a lot of these ideas with quantum, which at first I thought some might be quite abstract.

We're starting to synthesize and find ways to, to pull them together. And so I'm hoping that can be used now. I'm, I'm working with some embodied workshops and taking that into communities and letting this work inform how I coach, how I facilitate, how I, uh, try and explain.

So that's the hard bit, but, uh, I'll pass it back.

And it could, but shouldn't go without saying just massive appreciation to everybody who does and continues to show up for act inf lab. Um, you see the tip of the iceberg on the live streams, but there are many others.

So as we continue to formalize and determine and reduce our uncertainty about what kind of thing we are here, we'll hopefully have clearer and clearer ways to source attribution and to show recognition for all the people who contribute in just the million ways that are possible.

Um, from beginner to very familiar.

So we will just provide a brief overview of act inf lab from a narrative perspective before we jump into our strategy discussion.

So in 2020 towards the middle and in the end, some of us wrote a paper in, uh, September was when it was published on active inference and behavior engineering for teams, which was addressing a gap that we perceived in the literature, which was the relationship between active inference framework and remote teams and communication.

At the end of that year, after thinking about what our next steps would be after that paper, we thought that

something like an active lab would be cool.

So we sent out an initial call for collaboration during all of 2021 was our first year of active lab project operation.

And the projects were in the three organizational units of education, communication, and tools this year in 2022, which is what we're going to be talking about.

Today.

A lot has happened at both the lab scale and the project scale.

So we're going to first talk about the lab scale and the two pieces under lab scale updates are going to be the initial release of the lab strategy document.

And we'll also discuss the initial advisory board cohort.

And then we'll go into the project scale and discuss some of the ongoing projects that have resulted in all kinds of deliverables and outputs.

So first to the lab scale, Alex, would you like to talk a little bit about just from a high level strategy?

Alex Kuzminis, Yeah, thanks.

Alex Kuzminis, Yeah, thanks.

Alex Kuzminis, I think as everything in our lab, we are trying to do things in some new ways.

Alex Kuzminis, I think as a strategy that was not used before and also trying to put it on some state-of-the-art theories and approaches.

So as for strategy, we consider strategy not as a concrete document, but as an ongoing process, which will have quarterly reviews and updates.

And on high level, we are following open handedness approach.

So for who is interested, it's now can be put can be approached from developments of Kenneth Stanley.

And we are trying to avoid using classic objective missions, goals and our corporate stuff.

So we are trying to see in the future, we are trying to see some stepping stones that we can make next steps.

And also for sure, from the beginning, we use active inference not only as a research field, but also trying to implement and use concepts from where such as preferences and expectations.

And also we consider that all our activities are on different scales.

So we need to follow and understand how it's applied on different scales.

So thank you.

So we will now go through the current state of the strategy document and just anyone here feel free to raise your hands.

And of course, anyone watching along, feel free to write questions and we'll get to them.

So pause the video or check it out to read the full text here.

But just to summarize our strategic document preamble in active lab, we're directing our attention to different phenomena across multiple scales.

And the scales that we're most interested in or the ones that we are heuristically using are the personal, the individual human scale, the lab scale, that's act-inf lab scale, and then the community or the niche or the environment scale, which is the epistemic surroundings of the lab.

And what's special about the lab scale is that is where the lab can implement policy.

We can emit actions that are perceived as sense states by individual people or by the external community, and that may or may not change their policy selection.

But the lab scale is where strategy happens because that's where we can actually implement policy selection.

And we're taking this multi-scale nested systems perspective to five different aspects of act-inf lab.

And those five aspects that we're going to talk about in the next few slides are education, research, outreach and engagement, methods, and open-endedness.

Any general comments before we continue?

Great.

So act-inf lab is an educational nonprofit.

This plays out at all scales as all things do, but we're especially interested here in the personal scale in educating individuals.

In having individuals that learn and update their generative model.

And also respecting just the tremendous diversity of active inference journeys that different individuals are coming in and through relationship with the active inference lab.

And at the more community scale, our educational efforts are oriented towards reducing research debt and increasing the applicability and the

and the comprehensibility and the rigor and accessibility of active inference and related ideas.

Any comments on education?

Okay.

The second aspect of the lab is the research aspect.

And here it also plays out at the personal and at the community scale.

At the personal scale, we hope to provide a niche that scaffolds and catalyzes the research projects by participants, which is something we'll come back to later.

But we both do research in the lab as well as on the interface and just hope to motivate and create a space for people to share their individual research updates.

And at the community scale, we're recognizing that research is carried out alongside ongoing discussions about responsibility and ownership and many, many other terms.

So there we hope to learn practices and improve and give our own twist on what the future of science, open science, decentralized science, de-sci, just science with no adjectives and education.

What will research and education be in our future?

Steven?

Thanks, Daniel.

Yeah, this is really helpful to read.

And also the fact that we've got lots of people from different backgrounds coming together.

We were able to.

The research that's happening is quite transdisciplinary and it's been informed by lots of different ideas.

So there's a kind of a mixture of those processes.

So it's helpful to have the systems engineering work that was probably coming to later, but that's helpful in letting us get a feel for how the research doesn't get siloed as purely a paper.

But okay, how could that be bounced off to maybe make a difference in the world in ways that maybe people who don't even know active inference find that they're using something differently because active inference has helped the way that their practice is shaped.

So I like that.

Awesome.

The third aspect is outreach and engagement.

And the active inference lab is working on a global scale with a participatory model.

So at the personal scale, we welcome all participation and engagement.

And that means all time zones and spoken languages and preferences and what people want to do and where they're at with active inference participation in a sense begins with where the participant is.

And there's, of course, more that I hope everybody can say and add and show, not tell or show and tell if they've brought enough for the class in the coming years.

And then at the community scale, we also seek to be part of a emerging landscape of high quality science communication and participation and inclusion in research and projects around topics like active inference, which is what we're most attuned to in our regime of attention.

So that doesn't mean we only talk to active inference people.

So we hope that our outreach and engagement is meaningful and sustained and impactful.

Any comments on outreach and engagement?

Okay.

Methods.

The active inference lab applies state of the art methods across domains.

That refers to the methodology of active inference and the emerging tools and ways that we have of working with active inference, as well as other methodologies like for project management, for education, for communication, etc.

So at the personal scale, we're educating participants so that they can grow their expertise in different disciplines and also develop as a person holistically, like keeping it balanced.

You're not going to stay balanced.

You're not going to stay up and finish reading every single active paper.

So staying balanced and having methodologies that help us be performant, but also balanced are quite important.

And then the lab scale, we hope to implement the cutting edge, the Pareto optimal, the Bayes optimal methodology, which includes our switching strategies.

And we fix typos on the fly.

And we fix typos on the fly.

That's what editable documents allow.

Um, our affordances as an organization are the methodologies that we scaffold in projects.

So especially at the lab scale, we hope that we use methodology that's powerful and facilitates collective intelligence and also clarifies the distinction or the delineation, the partition, perhaps even between those who are following along and listening to live streams, reading the papers in the game.

That's awesome.

And then those who actually onboard into the lab and eventually have a role in a project will be able to increasingly be clear about what that means through the uses of different systems, ideas like practices, cases, responsibilities, and concerns.

Yes, Ivan.

Ivan Kuznicki- Yeah, thank you.

I would like to add that if we compile methods and outreach, so the lab provides a lot of opportunities for the understanding of active inference and we provide a lot of tools.

And for those who already participate in the lab and for those who just start to deep inside the active inference,

Ivan Kuznicki- Awesome.

Ivan Kuznicki- Awesome.

Ivan Kuznicki- And the last but not really the last aspect is open-endedness.

Ivan Kuznicki- As Alex mentioned earlier and has been bringing and reminding us all of since the beginning, that active lab is on the path of open-endedness.

Ivan Kuznicki- It doesn't mean that we sit and wait where the road forks or when we have uncertainty about the future.

Ivan Kuznicki- Those are fundamentals.

Ivan Kuznicki- And in fact, open-endedness is action amidst uncertainty.

Ivan Kuznicki- So at the personal scale, we recognize and support the open-endedness of the personal journey of individuals who are learning and applying active inference.

Ivan Kuznicki- So even if you don't know exactly where you're headed or what your finish line or your milestones, your checkpoint looks like, if you don't even know what your preferences are or you're unsure about some other parameter in your generative model, that's all good.

Ivan Kuznicki- At the lab scale, our strategy is to say it is a work in progress would be to almost assume that there would be a final strategy.

Ivan Kuznicki- But rather, our strategy is an ongoing process that will always be updating in response to changes in the niche and the effect that our actions have over multiple timescales.

Ivan Kuznicki- So we do recognize and utilize approaches to achieve short-term success and realizing products and deliverables over short terms.

Ivan Kuznicki- But also there's a time and a space and a place for thinking about active inference and the active inference lab in deeper time.

Ivan Kuznicki- And that is especially where diverse participation and our lab advisory board come into play, which we'll talk about in a second.

Ivan Kuznicki- And then just lastly at the community scale.

Ivan Kuznicki- It's true in principle that active lab can participate in various kinds of activities that might be beyond education, research and outreach and engagement.

Ivan Kuznicki- We will always be considering those opportunities in a case by case way and assessing whether it's something that fits best within the Markov blanket and the kind of thing that active lab is, or maybe some thing else could exist that could play that role in the ecosystem.

Ivan Kuznicki- Yes, Dave.

Ivan Kuznicki- Okay, Steven.

Steven Kuznicki- I'm interested by that.

Steven Kuznicki- That way that the ecosystem this this this this question of what's in the Markov blanket and the non equilibrium steady state and what's in the niche and we.

Steven Kuznicki- Those three things are kind of dynamic.

Steven Kuznicki- So I think over time.

Steven Kuznicki- Looking at the way the world is and without getting to evangelical there's a lots of big, you know, there's big claims made about active inference and its potential.

Steven Kuznicki- And as we've been having conversations we've seen a lot of things at scale come up, which is quite exciting conversations around that but often maybe a little bit fuzzy a little bit conceptual.

Steven Kuznicki- So I think.

Steven Kuznicki- Some of the stuff that's going to come out there the labs going to be spinning off as much as it might be holding on to.

Steven Kuznicki- Excellent.

Steven Kuznicki- Thank you.

Steven Kuznicki- So that concludes the discussion on the strategy update.

Steven Kuznicki- The second lab scale update is to mention our advisory board.

Steven Kuznicki- So this information can be found at our website active inference dot org and on the top right head over to where it says advisory board.

Steven Kuznicki- The advisory board is a insight and guidance board.

Steven Kuznicki- It's not a managerial board and it exists to help the lab make better strategic decisions.

Steven Kuznicki- So we engage our advisory board participants in terms of their expertise, different recommendations and questions they have for the lab.

Steven Kuznicki- And we recognize and hope to draw on and synergize with their experience in many, many different areas.

Steven Kuznicki- We have an image here from Kirchhoff et al. 2018 about nested systems and about the way that as systems develop, for example, as potentially a overdevelopmental time, a zygote goes from being a single cell enclosed by a lipid bilayer towards increasingly ramified levels of embryonic complexity or similar transitions could be happening over evolutionary time.

Steven Kuznicki- Something like that is occurring as we elaborate and build the ActInf lab.

Steven Kuznicki- We have an awesome initial cohort of the advisory board and it includes Bradley Alicia, John Boyk, Matt Brown, John Klippinger, Scott David, Jeff Emmett, Chris Fields, Carl Friston, Rafael Kaufman, Anatoly Levinchuk, Rosalyn Moran, Elba Serrano, Cheryl Vanhoof,

Steven Kuznicki- Tim Verbellen, Swan Webb, and Michael Zargom.

Steven Kuznicki- And we have already received really important and meaningful feedback from them and really appreciate it.

Steven Kuznicki- And for those who are interested in the kinds of work that we're up to and might have experience in a certain domain that could be helpful, for example, related to educational nonprofits, then perhaps get in touch with us.

Steven Kuznicki- But this is a major change to the lab scale to introduce an advisory board and we're off to a

fun start with them so far.

Steven Kuznicki- So we look forward to just seeing how that continues.

Steven Kuznicki- Any other comments on ActInf at the lab scale or we can head to the projects.

Steven Kuznicki- Okay, so as we head through these projects, feel free to raise your hand at any point if you want to add a comment or a question.

Steven Kuznicki- And as a participatory lab, those who are listening and you've made it this far recognize that all of these reflect the contributions of participants and reflect affordances for you to contribute further.

Steven Kuznicki- Even if only to reduce your uncertainty about the documentation or the way that something is occurring.

Steven Kuznicki- So hopefully if you're listening to this string of symbols, know that you can participate and perhaps you even should.

Steven Kuznicki- We're going to talk about a few different projects today.

Steven Kuznicki- In fact, 11 difference of them.

Steven Kuznicki- And you can see loosely on the left that these are in three organizational units, EDU, comms and tools.

Steven Kuznicki- But the project scale is really where the team and the roles and the practices occur.

Steven Kuznicki- So we're highlighting the project scale and the project is also the unit of granularity where you can really get involved and make a contribution.

Steven Kuznicki- So showing up for our synchronous organizational unit meetings is an awesome way to kind of paint a first coat and to be exposed to many of the projects.

Steven Kuznicki- But by and large, the contributions happen at the project scale and asynchronously.

Steven Kuznicki- So all backgrounds, all time zones, all levels of commitment.

Steven Kuznicki- Everyone is welcome to get involved with these projects, which we're going to go through now.

Steven Kuznicki- They're going to be the active inference ontology, the active inference course, internal educational initiatives like the systems thinking reading discussion group and ontology working group.

Steven Kuznicki- Active Inference Journal, research updates, live streams, podcasts, social media, active blockference, robotics and embodied, and active lab meta modeling.

Steven Kuznicki- On we go to the ontology.

Steven Kuznicki- We've just recently pushed an update to the active inference ontology and it's available at this Coda link here.

Steven Kuznicki- The goal of the active inference ontology is to increase the accessibility and impact of active inference by increasing clarity and interoperability and expressivity and all these other good things about terms and ideas.

Steven Kuznicki- And when you go to this site, you'll find that there's a explanatory preamble.

Steven Kuznicki- And what this ontology release contains primarily are two different kinds of things, proposed definitions and translations.

Steven Kuznicki- So first, the proposed definitions.

Steven Kuznicki- This is what the proposed definitions look like for the core terms.

Steven Kuznicki- The terms are on the left column here.

Steven Kuznicki- And then everywhere there's an at and a blue is like a clickable link to another term.

Steven Kuznicki- So it's kind of like how dictionaries use other words in that language.

Steven Kuznicki- And it will say A is kind of like B, except it's different than C.

Steven Kuznicki- And so that's what the definitions are.

Steven Kuznicki- They're natural language answers to the question, what is X?

Steven Kuznicki- And so these proposed definitions are currently in natural language, but there are other ways to develop them more formally.

Steven Kuznicki- The proposed definitions are there for those who just say, well, what does this term mean?

Steven Kuznicki- It also helps with computationally aided sense making.

Steven Kuznicki- And then the second piece that is in this current release of the ontology is the translations.

Steven Kuznicki- And so here's the current view of the top of the translations table.

Steven Kuznicki- So we have the term in the English on the left side.

Steven Kuznicki- And then we have so far complete translations of the core ontology terms into Spanish, Portuguese, French, Italian, Czech, Russian, and Hebrew, and some German and some others that are sort of beginning.

Steven Kuznicki- So especially if you have fluency or friends with fluency in some of these languages, but especially new and different languages, this is a really awesome opportunity to translate some terms, which are mostly normal natural language terms into another human natural language.

Steven Kuznicki- And that will facilitate the ability to, for example, deploy educational information or transcripts in their translations, do it in a way that's rigorous and accessible and meets people where they are in terms of their language fluency.

Steven Kuznicki- Yes, Dave?

Steven Kuznicki- Okay.

Steven Kuznicki- Okay, so that's the actin ontology project where we're developing a participatory ontology for active inference terms. Steven?

Steven Kuznicki- Could I just ask, not putting you on the spot, but if I was to ask you like what areas have been the most challenging or the most kind of like

Steven Kuznicki- Most most difficult to overcome and what what are the sort of questions that still being grappled with on the ontology, you know, and which ones have we overcome already and which ones are still sitting there in the background.

Steven Kuznicki- I'm just curious if you've got any thoughts on that.

Steven Kuznicki- Yeah, good question.

Steven Kuznicki- I'll give it thought and then anyone else is welcome to if they want.

Steven Kuznicki- One interesting thing is these definitions, although many papers do sometimes define their terms.

Steven Kuznicki- Either in the text or in a box or in a glossary.

Steven Kuznicki- That doesn't mean that the way that the terms are defined are coherent or compatible or consistent.

Steven Kuznicki- And a lot of times it's like a yes and someone defines it one way and someone else just kind of takes a different spin on it.

Steven Kuznicki- But other times we have come across several situations all together, all of us at least here, where there were incompatible seeming uses of terms.

Steven Kuznicki- And perhaps that happens with all kinds of linguistic elements.

Steven Kuznicki- So that was one interesting thing was that it's not just like we're looking through papers and copying sentences.

Steven Kuznicki- There's actually a really powerful process of the discussion that we had focusing week by week on just a few terms or one term.

Steven Kuznicki- Where we were seeing all the different ways that different terms are being used inside and outside of active inference and just trying to work towards some kind of grip on that blue.

Steven Kuznicki- So one of the biggest challenges that I think we face is dirty and incomplete data.

Steven Kuznicki- Which is like pervasive across fields and disciplines everywhere.

Steven Kuznicki- So I mean, trying to be the bridge to fill the gaps to make the data to complete the data like and I mean, there's always opportunities that people want to get involved.

Steven Kuznicki- Like in filling out the data like helping us to complete, you know, to have a complete picture of which to meta model.

Steven Kuznicki- Awesome. And there are other columns and features of the ontology that we're working on in the lab for anybody who wants to get involved like correct examples, just simple, declarative, concise, positive examples of terms being used correctly.

Steven Kuznicki- That's awesome.

Steven Kuznicki- Incorrect examples.

Steven Kuznicki- People who are using sentences.

Steven Kuznicki- It doesn't have to be caught in the wild, but it could be.

Steven Kuznicki- But just generating sentences that are like counter examples can be very helpful for both humans and computers in their learning.

Steven Kuznicki- And so we have examples, counter examples, formalisms, connecting core terms and other terms to equations and papers and connecting it with the literature corpus.

Steven Kuznicki- So there's so many angles.

Steven Kuznicki- Those with curiosity and expertise in active inference and or ontology should absolutely get involved with this, which will be a long term project of the lab and we hope to find new ways of deploying it.

Steven Kuznicki- And just to give one little contextualizing slide here.

Steven Kuznicki- This entire continuum can be considered to be different types or grades or like a taxonomy of ontologies.

Steven Kuznicki- And so ontologies in their most informal can be as simple as a list of terms or glossaries, the sources, or what we have right now with the current release, which is like a principled informal hierarchy and some relationships amongst terms.

Steven Kuznicki- And that's what we have right now with the proposed definitions and translations.

Steven Kuznicki- We're also working with the ontology working group, which we'll talk about in a few minutes on formalizing that ontology, for example, using Sumo and SUO-KIF and thinking about how do we integrate the ontology with course development, with knowledge engineering and literature integration.

Steven Kuznicki- So some of the many things that people can look forward to.

Steven Kuznicki- Okay, on to the second project, which is the ActInf course.

Steven Kuznicki- So in the active course project, we are seeking to create a learning platform that embodies the principles of active inference and facilitates the learning of its principles and concepts to learners of many different backgrounds.

Steven Kuznicki- And this is something that we are not too far down the road on, but are eagerly looking forward to, especially in light of some recent epistemic changes in our niche, which we'll talk about soon.

Steven Kuznicki- But there's a massive potential for active inference education.

Steven Kuznicki- And we hope to have course offerings and educational experiences that meet that opportunity.

Alex?

Alex Kuznicki- Yeah, just a quick note to avoid possible misunderstanding that notion platform here is not like for some IT platform, but more about set of educational services, which will be supported to improve learning affordances.

Steven Kuznicki- Yes, good call thanks for clarifying that.

Steven Kuznicki- So if you're interested or curious about course design education education in a participatory way nonprofits.

Steven Kuznicki- Those are all excellent cues for people who would like to come on board and help in a serious way for this course as we start to think about what it can be.

Steven Kuznicki- Okay.

Steven Kuznicki- Third project.

Steven Kuznicki- We have several internal educational initiatives in ActInfLab.

Steven Kuznicki- And so on the left, we see the systems thinking reading discussion group, and on the right, ontology working group, the OWG.

Steven Kuznicki- Alex or anyone would you like to describe or give a note on either of these sides?

Steven Kuznicki- I just want to say that from very beginning we consider that we should not only learn and educate active inference, but also to learn, as you mentioned before, state of the art practices.

Steven Kuznicki- And system thinking was one of the main discipline from the beginning, what we think is very useful for collective work, especially in remote mode.

Steven Kuznicki- So it's great that now more and more people start learning and it definitely will improve our communication and speed up our activities on the projects.

Steven Kuznicki- Thank you, yes.

Steven Kuznicki- Yvonne has been facilitating the systems thinking reading discussion group STRDG.

Steven Kuznicki- And we have an intrepid group of participants who are working their way through this online course and having regular discussions and asynchronous conversations as well around it.

Steven Kuznicki- And as Alex said, this is really important for the ways that we work together and communicate.

Steven Kuznicki- And on the right side, we have the ontology working group.

Steven Kuznicki- So previously we were discussing the act-inf ontology as a project where we're talking about active inference terms and how they're related.

Steven Kuznicki- In the ontology working group, we are increasing our competency in computational

ontology.

Steven Kuznicki- And what that has looked like primarily has been working with the SUMO suggested upper merged ontology system in SUO-KIF and working through the textbook of Adam Peace.

Steven Kuznicki- And this has been an awesome experience. Dave has made a massive amount of contributions and ontological limericks of different formats.

Steven Kuznicki- And that has also been a great learning experience that hopefully will fold back in towards improving the active inference ontology, but it's also just a learning initiative.

Steven Kuznicki- And that brings us to an ongoing learning initiative.

Steven Kuznicki- Which is the release literally in the last several days of the textbook by Thomas Parr, Giovanni Pizzullo, and Carl Friston.

Steven Kuznicki- It's the active inference textbook.

Steven Kuznicki- And we asked, well, what can the act-inf lab do for slash with around the textbook?

Steven Kuznicki- Do you want to join a textbook group?

Steven Kuznicki- Not just a reading group.

Steven Kuznicki- Just a textbook group.

Steven Kuznicki- Because we are going to be organizing cohorts of textbook groups.

Steven Kuznicki- And you can email us for the form if you don't already have access to it.

Steven Kuznicki- But you will fill out a form that asks whether you'd like to join an upcoming textbook group cohort.

Steven Kuznicki- And it will be done in groups and cohorts.

Steven Kuznicki- Yes, you're ready or no, you'd like to perhaps hear about that affordance in a few months.

Steven Kuznicki- We ask for name, email address, and country, whether you have a textbook or not.

Steven Kuznicki- And then just for you to provide several sentences or more on your level of familiarity and your interest in active inference.

Steven Kuznicki- And all backgrounds and familiarities, as with all activities, are welcome.

Steven Kuznicki- This is going to be a really awesome experience, I think, to work through this textbook and see it as just one pebble on the anthill or the tip of the iceberg or any other metaphor that people use for the way in which this textbook will both be a visible and centralizing artifact,

Steven Kuznicki- As well as something that is really only a few hundred pages long and it's a relatively small physical book.

Steven Kuznicki- So it's just the beginning, it's not the end of active inference.

Steven Kuznicki- And those in the textbook groups will realize that and enact it.

Steven Kuznicki- Any comments on internal educational initiatives?

Steven Kuznicki- All right.

Steven Kuznicki- Number four.

Steven Kuznicki- The Active Inference Journal.

Steven Kuznicki- So in this project, we are publishing knowledge artifacts so that they are searchable, indexable, and citable.

Steven Kuznicki- Most recently in the Active Inference Journal, we uploaded a second improved version of

the transcript of the Carl Fristen Symposium, Applied Active Inference Symposium, that was in June 2021.

Steven Kuznicki- And from that, we are improving our methodology that we learnt and learnt by doing in that first Symposium transcript process and applying that to other event recordings.

Steven Kuznicki- For example, Guest Stream 16 with Mark Solms and Live Stream Number 40 with a quantum free energy principle.

Steven Kuznicki- So the journal is going to be a great place where we continue to build a participatory pipeline for improving spoken and written dissemination.

Steven Kuznicki- Anything that anyone wants to add on this?

Steven Kuznicki- Yes, Dave.

Steven Kuznicki- Yeah, I sent the original paper and the three video segments to Sean Carroll along with a summary of the three big points in the summary.

Steven Kuznicki- He says, this is cool.

Steven Kuznicki- I will check it out.

Steven Kuznicki- Which I'm sure incentivizes the gentleman who volunteered to go through our transcript and make it more presentable.

Steven Kuznicki- To get through that and then we can send that to the very busy Dr. Carol who just has a new appointment.

Steven Kuznicki- He is something like the the chair of the philosophy of science at Johns Hopkins.

Steven Kuznicki- So all his pain of being denied tenure is paying off.

Steven Kuznicki- And he may pay off.

Steven Kuznicki- He already I think gets along very well with Professor Friston.

Steven Kuznicki- He, he says, I have never had to work so hard in an interview with somebody who is so knowledgeable and well spoken and anxious to communicate.

Steven Kuznicki- Awesome.

Steven Kuznicki- Yeah.

Steven Kuznicki- And it's a great example of a project that increases accessibility.

Steven Kuznicki- For example, by taking two or six hours of YouTube discussions and making them searchable and translatable.

Steven Kuznicki- And then also reduces research debt because we can have somebody spend three or seven hours improving and curating a manuscript and finding, oh, when somebody mentioned this paper, here's that specific citation.

Steven Kuznicki- That multiplied by thousands of people searching for that paper, maybe they have access, maybe they don't, maybe they misheard the name.

Steven Kuznicki- There's so much that we can be proactive about in terms of making our knowledge corpus really accessible and meaningful and woven according to modern practices.

Steven Kuznicki- Okay.

Steven Kuznicki- Awesome work though, Dave and others on the journal, Maria and others as well.

Steven Kuznicki- Okay, on five, we have research updates.

Steven Kuznicki- So this is not a project that we actively steward per se in the lab, but this is just capturing research that's in progress and finished by lab participants.

Steven Kuznicki- So just to highlight a few papers and products that have been released by active inference lab participants, but not necessarily as an active lab project in the last few months.

Steven Kuznicki- So by Avel, Gwen and Carlou, AKA Serval, AKA Icarus, the paper Cognitive Agency in Sociocultural Evolution.

Steven Kuznicki- And then an awesome collaborative paper that we wrote with myself, Sean, Arhan, RJ, Shadi, Avel, Blue, Yvonne, Sid, Amit, Jakob, Caleb, and Alex, where we talked about decentralized science, DSCI.

Steven Kuznicki- We integrated some of the more technological perspectives on DSCI, for example, related to Web3 and blockchain, and juxtaposed that with some of the bottom up, the view from the bottom participatory sense making and epistemic commons, and produced an active entity ontology for science, EOS, that we hope will allow us to have a way to model some of these complex epistemic ecosystems in the coming months.

Steven Kuznicki- So that's just some of the research and there's other awesome research that's in progress.

Steven Kuznicki- Okay, on to yes, Steven go for it.

Steven Kuznicki- Yeah, I'd be curious, I think this work with situated sense making that decentralized science is really, really interesting.

Steven Kuznicki- And I think there'd be ways that the other areas of sense making which this can be helpful to be able to do.

Steven Kuznicki- And I think that active inference, generative modeling approach with other ways.

Steven Kuznicki- So I'm really glad that you did this work actually as a group, and I think it's going to help tell another story which might have been a bit too hard to tell from the ground up, but now you've got a bridge.

Steven Kuznicki- So thanks.

Steven Kuznicki- Cool.

Steven Kuznicki- Okay, on to six.

Steven Kuznicki- So we had awesome live streams as usual in this quarter.

Steven Kuznicki- At the coda.io slash active inference lab active lab, you can find the past and upcoming live streams.

Steven Kuznicki- We had several guest streams and other kinds of streams, but most of our streams in the last few months were the paper discussions.

Steven Kuznicki- So in live stream number 35.0, one and two.

Steven Kuznicki- So as with all of these, there was a dot zero, a background and contextualizing video that was prepared in the weeks leading up to the dot one dot two discussions, which are participatory group discussions.

Steven Kuznicki- So if you ever see a paper where you want to be in the dot one or the dot two, and be part of a group and you don't need to say a ton, no one's going to call on you.

Steven Kuznicki- You can ask anything you want, provide any perspectives.

Steven Kuznicki- Always welcome to join that way.

Steven Kuznicki- And then something that's incredibly helpful for the lab is to commit assuredly ahead of time and contribute to the dot zero, making the slides and helping contextualize the paper ahead of the dot one and the dot two.

Steven Kuznicki- So we, in the last few months, have talked about in number 35, we talked about a tale of two architectures, free energy, its models, and modularity.

Steven Kuznicki- Check out the live streams to learn more. I won't even attempt to summarize the papers other than just to state their titles.

Steven Kuznicki- In 36, we discussed modeling ourselves, what the free energy principle reveals about our implicit notions of representation.

Steven Kuznicki- We then dove a bit into the technical in 37 with free energy, a user's guide.

Steven Kuznicki- In 38, discussed the evolution of brain architectures for predictive coding and active inference.

Steven Kuznicki- In 39, discussed morphogenesis as Bayesian inference, a variational approach to pattern formation and control in complex biological systems.

Steven Kuznicki- And then in 40, we discussed a free energy principle for generic quantum systems, which was, as anybody who was there before or during or after it happened can attest, was quite a special and interesting experience.

Steven Kuznicki- And just to show what that CODA page looks like, if you head over there, you'll see the upcoming streams.

Steven Kuznicki- Here we are in the end of March. It's not in the live stream section. It's in the round table section. Here we are right now.

Steven Kuznicki- But in live streams, you'll see the papers that we're going to be discussing in the coming months.

Steven Kuznicki- So if you're interested in any of these upcoming papers, please get involved in the .0 or the .1 and 2.

Steven Kuznicki- If you're interested in inviting anybody or yourself being a guest on a guest stream, that's always in affordance.

Steven Kuznicki- These are like one off or small series that can be on active inference research or related topics.

Steven Kuznicki- And we've had some great discussions in the last few weeks on guest streams.

Steven Kuznicki- We have model streams and math streams that are for presenting some of the formal and model based components of active inference.

Steven Kuznicki- And org streams where we talk about organizational principles and organizational design.

Steven Kuznicki- So those are just some of the live streams that we've had and it's been fun.

Steven Kuznicki- So especially if you want to give a guest stream or if you can connect to somebody who does want to or encourage or recommend somebody or you want to be a facilitator for someone else to present or any other combination of affordances.

Steven Kuznicki- Let's make it happen because we can get a lot of awesome modifications in our niche.

Steven Kuznicki- I think what they used to call making content back in the web two days.

Steven Kuznicki- So that's all for live streams.

Steven Kuznicki- Podcast.

Steven Kuznicki- Blue, do you want to give an update?

Steven Kuznicki- Sure.

Steven Kuznicki- So we are moving through the podcast really revisits a lot of the early live streams.

Steven Kuznicki- So like if you're playing catch up like if you caught a live stream, you're like, Oh, I should go through the back catalog.

Steven Kuznicki- Like maybe now you're on live stream 20 or something like that.

Steven Kuznicki- The podcast starts with the early episodes and I think we're through episode nine live stream number nine now.

Steven Kuznicki- So yeah, check it out.

Steven Kuznicki- It's fun.

Steven Kuznicki- Also a great chance to get involved if you have like a V expertise would love to get some pointers also like I know some stuff i've been clipping the podcast for a little bit so.

Steven Kuznicki- If you'd like to learn like the basic things that I know and want to get involved in making the podcast reach out, I could always use help if you want to listen to more podcasts come help me make them.

Steven Kuznicki- Dave.

Steven Kuznicki- Yeah, if you have listed the podcast after nine that you're planning to do i'd like to get that because I want to get started with just doing basic text conversion on this if that assumes that this, these are the priority is that correct.

Steven Kuznicki- These are the ones that you that we want to get in front of you.

Steven Kuznicki- You know, I don't know what the what the priority is there, Dave, but I will say like there's a lot of things that you can get involved.

Steven Kuznicki- But I will say like there's definitely value to having a transcript.

Steven Kuznicki- I actually looked at pod bean who is our podcast platform the other day, and they will transcribe our podcast for us actually at 20 cents a minute.

Steven Kuznicki- So if you really want to know what the value of transcription is just just the value to just put it into text now I don't know if that comes with periods or punctuation or I don't know what that output looks like.

Steven Kuznicki- But there's definitely value to having a transcript there.

Steven Kuznicki- I'm not sure that these are necessarily the priority but it's just going through the back catalog it's helping me catch up.

Steven Kuznicki- So I just kind of started where I wanted to start at the beginning when I started the beginning right I don't know.

Steven Kuznicki- What what I love about these podcasts is they're short.

Steven Kuznicki- They're like from four to maybe 15 or 10 minutes long.

Steven Kuznicki- And the first time that a speaker speaks, they introduce themselves.

Steven Kuznicki- And there have been some probable clippings and rearrangements which are very subtle that help that listenable piece be like coherent like it's like an extended thought even one that's distributed across multiple speakers.

Steven Kuznicki- So like you hit play listen to it and then you can think and reflect because sometimes when it's two hours on a paper, it feels like it's like a rail car coming through where it's like every single idea and thing and people say that they pause the live streams and there's so much.

Steven Kuznicki- And then also as the priorities.

Steven Kuznicki- That's something we can figure out and discuss.

Steven Kuznicki- But won't it be amazing when the live stream is you know the front car of that train piercing into the future hashtag doing it live.

Steven Kuznicki- And then there's transcripts for all the live streams and that can be on one hand connected to the ontology.

Steven Kuznicki- A transcript of the live stream itself could exist.

Steven Kuznicki- There could be some clipping and some transcripts for the podcast.

Steven Kuznicki- So as we develop our informational ecology.

Steven Kuznicki- All of these kinds of connections and tasks are in play.

Steven Kuznicki- It's just a question of how late can you stay up, which I've wondered about for a long time.

Steven Kuznicki- Dave needs no sleep.

Steven Kuznicki- You know, the other cool thing about the podcast is that it's slide free.

Steven Kuznicki- So where when you're on the YouTube watching the live stream, often it's associated with some visual image.

Steven Kuznicki- And sometimes there will be like a description of the image in the podcast, but it's nothing that you need to like pull over and look at.

Steven Kuznicki- Which is sometimes if you're trying to like watch the YouTube videos on a long drive.

Steven Kuznicki- It's like, oh, that might not be safe.

Steven Kuznicki- But the podcasts are definitely, you know, commute friendly because you don't have to see anything.

Steven Kuznicki- You can just listen.

Steven Kuznicki- So that's another it just offloads that the visual input as well.

Steven Kuznicki- Good for the treadmill stuff like that.

Steven Kuznicki- Great point.

Steven Kuznicki- Okay, on to eight.

Steven Kuznicki- So some updates on our, I'll just call it social media.

Steven Kuznicki- We have our Twitter with 1,359 followers.

Steven Kuznicki- We have about, I think we crossed 300 members in the discord.

Steven Kuznicki- Over 770 subscribers on YouTube and some several hundreds of people reached in Facebook.

Steven Kuznicki- And anyone who is interested in automating social media, that would be an excellent affordance.

Steven Kuznicki- Blue?

Steven Kuznicki- Yeah, definitely.

Steven Kuznicki- I was just you took the words right out of my mouth.

Steven Kuznicki- And Facebook is very new.

Steven Kuznicki- So we have like just 40 likes on our Facebook page.

Steven Kuznicki- So if you're listening to this, and you use Facebook regularly for whatever, I mean, it will ping you to remind you about podcasts and sometimes live streams.

Steven Kuznicki- Sometimes I say because like I said, we need an automator.

Steven Kuznicki- So that's your expertise and you want to always be reminded about the live streams.

Steven Kuznicki- Get in touch.

Steven Kuznicki- Yes, just to be able to preload a buffer.

Steven Kuznicki- Here in the front car of the train.

Steven Kuznicki- It's sometimes hard to clean up the pieces.

Steven Kuznicki- And so if you're feeling like that's something that's interesting to you, then there's many opportunities to have engagements on different platforms.

Steven Kuznicki- And just feel free to just engage in the comments and in the threads, however, makes sense.

Steven Kuznicki- Great enough with social media.

Steven Kuznicki- Okay, on to nine.

Steven Kuznicki- So now we're heading more towards the tools part of this discussion.

Steven Kuznicki- Primarily, the work of Jakob Smickel has been just excellent on active block framework.

Steven Kuznicki- Which is a package that we're developing in order to facilitate cognitive agent modeling.

Steven Kuznicki- And if you look up active block front on GitHub, you'll find the open source repository.

Steven Kuznicki- What we've done with active block front is implement active inference entities, the kind that you will see graphically described in papers or the kind that have been implemented in Python via PyMDP.

Steven Kuznicki- And we built upon PyMDP as well as a complex systems modeling framework called CAD CAD.

Steven Kuznicki- And by integrating the active kernel of the entity action perception loop with the complex systems modeling framework, we hope to have engineering grade approaches towards modeling complex cognitive ecosystems.

Steven Kuznicki- So if you are interested in Python or Julia or other aspects that are computational related to active inference and you like complexity and complex adaptive systems and everything else that you might see on this slide.

Steven Kuznicki- Active block front is an open source project.

Steven Kuznicki- So go ahead and fork it, make your contributions and come get in touch with us if you'd like to be a little bit more in the loop and really looking forward to seeing how this project develops in the coming years.

Steven Kuznicki- In dot tools also, we've been having some awesome discussions around robotics and embodied cognitive entities.

Steven Kuznicki- So we have been characterizing kind of like a literature review on different cognitive agent modeling frameworks.

Steven Kuznicki- So active inference is often proposed as a unified cognitive entity model.

Steven Kuznicki- And so we've been just curating many other cognitive agent modeling frameworks from the past and present, and then assessing them in terms of their language, what cognitive features they implement,

Steven Kuznicki- Whether they deal with embodiment, what is their mathematical underpinnings, and so on.

Steven Kuznicki- And meanwhile, characterizing some of those cognitive features and elements like anticipation, memory, counterfactuals, abductive reasoning, different kinds of cognitive features or motifs, and

then trying to understand how do these different cognitive model frameworks line up on the rubric of wanting to have expressivity across all of the

Steven Kuznicki- All of these different cognitive features and to integrate them.

Steven Kuznicki- And that's a little bit more on the symbolic cognitive.

Steven Kuznicki- We've also been thinking a little bit about the embodied, whether the embodied and spatial and bodily or the embodied and robotics.

Steven Kuznicki- So here is a screenshot from the work of JF Cloutier, who's been an awesome participant in this past quarter, and some active robots that are Carl and Andy.

Steven Kuznicki- I think people can probably guess what their last names are.

Steven Kuznicki- And there isn't too much more to add here other than check out this woefully under acknowledged channel and work.

Steven Kuznicki- And if you're interested in any of these areas related to cognitive modeling and robotics and embodied cognitive systems, those are the kinds of things that will be awesome to scaffold in the lab.

Steven Kuznicki- Okay.

Steven Kuznicki- And to our final of the 11 projects, Actimflab Metamodeling.

Steven Kuznicki- So Actimflab Metamodeling, AKA Alma.

Steven Kuznicki- So there's an A and an L and there's an M and there's an A.

Steven Kuznicki- And I think that was Jessica's suggestion, because Alma has nice connotations as well.

Steven Kuznicki- This project is using metamodeling tools like Coda to provide the capacity for the

Steven Kuznicki- To provide the capacity for lab scale, generative modeling, updates, and reconfiguring.

Steven Kuznicki- So the ability for the lab to reflect and to modify itself and its niche.

Steven Kuznicki- In order to design the kind of participatory niche that facilitates successful policy selection by lab members.

Steven Kuznicki- Like what does the lab need to provide the individual as the action state of the lab, sense state of the individual in terms of the communication such that that participant is chill, but also making the right decisions that are productive.

Steven Kuznicki- And we're using primarily Coda.

Steven Kuznicki- That's why the font is bigger here.

Steven Kuznicki- But we're also implementing the active entity ontology for science that we discussed on the research slide and active blockference just two slides ago as a more formal way to get at some of these metamodeling.

Steven Kuznicki- And this is a figure from our 2020 paper that first early paper on systems engineering and active inference and remote teams, where we also looked at the informational niche, the shared informational niche of the team.

Steven Kuznicki- This is their shared folder, their shared code document, their formal documents, their drive, the chat threads.

Steven Kuznicki- That's the shared epistemic niche, where actions to modify the niche by one participant can then be like stigmargy, modifying the environment so that themselves later or other participants then or in the future take different action selection as mediated by the way that that sensory state updates their generative model.

Steven Kuznicki- And then we have narrative.

Steven Kuznicki- And so we've been very interested for a long time in rhetorical and narrative approaches to cognition.

Steven Kuznicki- And so now we're just beginning to see how some of these approaches can come into play in the context of active inference and in the lab.

Steven Kuznicki- So that's like the meta on ActInfLab.

Steven Kuznicki- And this is also an awesome project set of affordances.

Steven Kuznicki- If you like CODA, if you liked Douglas Hofstetter's Gerdel Escher-Bach, if you like ActInfersServe, all of these would be great diagnostic cues that perhaps you would like active lab meta modeling.

Steven Kuznicki- Yes, Steven.

Steven Kuznicki- Yeah, thanks for sharing.

Steven Kuznicki- I was wondering how this also might even be able to tie in in the future to micro phenomenological kind of modeling.

Steven Kuznicki- I that that kind of approach. So we've got the meta, the narrative, the story, and hopefully maybe some sort of micro phenomenological approaches as well.

Steven Kuznicki- But I think we've got to walk, but we'll get there. So thanks.

Steven Kuznicki- When you said micro phenomenological, I thought, oh, like the experiences of children.

Steven Kuznicki- Yeah.

Steven Kuznicki- Say hello.

Steven Kuznicki- Cool.

Steven Kuznicki- So that is it for the project.

Steven Kuznicki- The project updates. They're all ongoing. Let's just kind of go over some next steps.

Steven Kuznicki- At the lab scale, several areas that we're especially looking for assistance or contributions on would be on the legal side.

Steven Kuznicki- For example, potentially connecting with a firm that would be interested in providing pro bono assistance, who has experience with education, nonprofits, intellectual property, tech, etc.

Steven Kuznicki- And also helping our lab navigate a course towards financial sustainability, whether through grants or other mechanisms.

Steven Kuznicki- But just finding a way that we can reward and stabilize the contributions of participants.

Steven Kuznicki- And then at the project scale, any and all participants who want to be in the strange loop, in the strange attractor, want to be on the path towards contributing to active inference lab, as they also grow and develop as an active inferencer and as a person.

Steven Kuznicki- So if any of these projects or some other opportunity that you see and want to take the initiative on any of these reflect affordances for you.

Steven Kuznicki- So hopefully if you're listening this far in a minute and six hours in, or wherever it's clipped out in a podcast, then just recognize that these are all reflecting the past contributions of participants.

Steven Kuznicki- Like yourself.

Steven Kuznicki- And this is just an opening.

Steven Kuznicki- It's only the first quarter of 2022.

Steven Kuznicki- We're just beginning.

Steven Kuznicki- And so there's so much in the field and in the lab to be done that there will be something to do that's meaningful and chill and aligned with your preferences and expectations and availability.

Steven Kuznicki- Just believe it.

Steven Kuznicki- Okay.

Steven Kuznicki- So that takes us close to the end and a little modification of our common ending slide.

Steven Kuznicki- Where do we go from here?

Steven Kuznicki- What are people still curious about learning and applying and what are we going to do differently maybe in the next quarter?

Steven Kuznicki- Yes, blue.

Steven Kuznicki- Yes, blue.

Steven Kuznicki- So I think I already put it out there at the beginning.

Steven Kuznicki- But I'm really like interested in not talking about things anymore spending less time talking about things and more time actually doing things so that that's what I'm going to do differently in the next quarter.

Steven Kuznicki- And that's me as like on a personal scale has nothing to do really with the lab, but it's I just want to just work now.

Steven Kuznicki- I just want to put my head in the books and get to it.

Steven Kuznicki- Awesome.

Steven Kuznicki- Yeah, sometimes like it helps to get caught up through conversation and just being in the synchronous meetings.

Steven Kuznicki- But one hour per week is not how we build a course or not how we go through a textbook.

Steven Kuznicki- And so that is a very important balance to keep in mind Steven or Dave.

any other thoughts yes or no there's a lot of good work going on i'd say just keep doing it and talk about you know talk about what's working well what needs to be improved what the bottlenecks are we've mentioned some bottlenecks and i know sometimes uh maybe some of the fun projects that could be pushed back in favor of some of the more difficult things that actually will end up being leveraged heavily the i think of the ontology it may not be fun to slog through this stuff but it's going to really pay off when it gets some transitive closure aspects of that of course awesome for those who are studying or curious about active inference but feel kind of like on their own whether no one else in their local niche or network or lab or university or organization is working on active inference that's like one of the most powerful things is just knowing like if you have an idea that you're not even going to pursue but just somebody should do this or it would be cool if this existed having a space where you can add that pebble and maybe down the road or maybe sooner than you expect somebody else seeing that pebble is stimulated to act and so that's what can happen when we have shared ways of working and that's just what we're beginning to see steven yeah sort of bouncing off that as well i think i keep thinking about stages of readiness which i i am i'm normally working with community participants who are marginalized but i kind of i've had that myself of like feeling a bit marginalized feeling am i

like ready and it was like a little jump but it can be a chasm you know trying to get to that stage and i think that can be for a lot of people you don't quite know on different things whether you're near that threshold or not and you're often a lot closer to being ready you maybe just need someone to give you a kick up the bum and say let's go for it you know but having said that that's when you appreciate the quality of getting getting some things under your belt on the way to the line so i think like blue said like it's time to act and um for me personally yeah it's like um the stage of readiness i'm at is very different to where i was a year ago for sure um both in my knowledge when i look back to where i was but also in my heart and that's a that's not an insignificant piece of the equation so thanks awesome i'll just wait one more minute if anyone who's watching live has any questions that they would like to see addressed but totally agree it's um it's a multi-year journey i mean we're all on a lifelong journey of some sort but hopefully even our active inference journey is multi-year and so being able to connect our policy selection our behavioral decisions in the next micro cicade of the eyes the next place your eye is going to look the next action you're going to take in terms of opening up a hyperlink or reaching for a book or a pencil but being able to connect those very rapid actions with a bigger picture and with the longer temporal time scale and acting amidst uncertainty that's what active inference is about one aspect of it and so if anything that you've heard resonates just hopefully see this roundtable presentation as like an invitation less than a stand and deliver because there's so much openness in the area and in the lab and it maybe can feel like one is starting late but that's just human knowledge development it's a treadmill that's been going and through coming together in the lab we can actually leverage each other's contributions and catch up on fields that have thousands of hours of reading and hopefully get somewhere cool together so um dave blue steven and ivan and alex from earlier and all the other lab participants big big thanks for this amazing quarter and it's been a fast one and on to the next one okay bye thanks all have fun thank you